for extending maximum human life span. The Walford book, although written by an eminent gerontologist, is unabashedly intended for the lay reader. Walford points out that if we succeed in conquering the late-life diseases and limit ourselves to "rectangularizing technologies," we are headed for insurmountable social and economic problems. However, by substantially prolonging the human life span, we would succeed in greatly postponing the onset of the major diseases of the aged, and this would be "perhaps a cheaper and better way to cure the killer diseases than trying to pick them off one by one, as medicine is trying to do at present." Less convincing is Walford's argument that such a prolongation would result in a decrease in the percentage of the total population that is senile or debilitated rather than just a postponement of the diseases of aging, and perhaps he paints too rosy a picture of the societal benefits of an extended life span.

Walford's optimism is based on studies providing evidence that the genetic program for aging "is written in only a few (regulatory) genes or gene systems," thus markedly reducing the number of systems that would have to be manipulated to increase the maximum life span. Specifically, he regards DNA repair as one such key system, and the level of repair rate as genetically programmed. He sees the immune system as equally important, since the normal immune responses decline with advancing age, and an array of self-destructive abnormal (autoimmune) reactions increase; the integrity of the immune system is controlled by the MHC gene system. Thus, Walford believes, by improving the genetic control of these two systems, we can extend longevity.

Both authors emphasize that we possess genetically programmed compensatory or corrective mechanisms and that longevity is limited by the rate at which these mechanisms age. Frolkis makes only passing reference to the role of the DNA repair and immune systems. Instead, he devotes his book to a comprehensive and detailed account of aging-related changes, both deteriorative and compensatory, in virtually every element of the neuroendocrine system. To express this concept of life-maintaining compensatory mechanisms, Frolkis has invented the word "vitauct" (from the Latin *vita* for life and *auctum* for prolonging).

Despite their differences in emphasis on systems important to life prolongation, both authors present similar strategies. These are based on experimental studies in animals that show a 30 to 50 per cent increase in maximum life span by institution of diets that provide what Walford calls "undernutrition without malnutrition." The strategies are also based on studies that show similar effects when body temperature is lowered. Walford emphasizes the results of these studies in terms of the DNA repair and immune systems, whereas Frolkis details their results with respect to the neuroendocrine system.

The Walford book provides easy and entertaining reading. The author's lengthy account of his personal dietary habits (including some 27 pages devoted to diets, recipes, and charts of the nutritive value of various foods), to demonstrate the immediate use of longevity-extending measures, does not improve his book. The Frolkis book makes for much more difficult reading. Its style would have been improved if an American had edited the translation. It is seriously flawed by many malapropisms and by pages that are out of numerical order. Nevertheless, it provides a comprehensive review of both Western and Russian studies on aging of the neuroendocrine system.

Both authors make a distinction between biologic aging and the late-life diseases. Walford acknowledges that these diseases are prevalent in patients with accelerated aging, and maintains that by postponing biologic aging, we would postpone the onset of these diseases. This suggests a more intimate relation between biologic and pathologic aging than is generally recognized. If biologic and pathologic aging are not readily separable, it may turn out that death in advanced old age will not be accidental, as Walford believes it will. I agree that postponing biologic aging may be a better strategy than trying to pick off each disease one at a time, and that a greater investment should be made in this strategy, whether it results in only rectangularizing the survival curve or actually extending it.

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## The Cancer Prevention Diet: Michio Kushi's Nutritional Blueprint for the Relief and Prevention of Disease

By Michio Kushi. 460 pp. New York, St. Martin's Press, 1983. $14.95.

Physicians and their teams, especially in the area of oncology, need to become aware of the unsound dietary advice emanating from "nutritionists" who are sensationalizing the National Academy of Sciences' recent report on diet and cancer. Patients with cancer who postpone or discontinue proper medical attention may do so in favor of unconventional and unsound "diet cures." Books like this one can serve only to encourage such dangerous practices.

Subtitled a "Blueprint for the Relief and Prevention of Disease," Kushi's book is based on his own theory, advanced without supporting data, that the macrobiotic diet is the key to the prevention and cure of degenerative diseases, including cancer. The macrobiotic diet is a strict vegetarian diet that allows no meat, poultry, eggs, or milk products but includes an occasional serving of white-meat fish or seafood. Fruit is consumed only once or twice weekly and is eaten cooked or dried. Tropical and semitropical fruits are taboo, including oranges, grapefruit, and bananas. The other forbidden items are too numerous to list but include all common sweeteners, most vegetable oils, potatoes, tomatoes, and "all industrially mass-produced foods."

The basis of the cancer-prevention diet theory is the traditional Asian yin–yang philosophy: everything in life is yin, yang, or in between. Harmony is achieved by balancing the two extremes. The macrobiotic diet supposedly represents such a balance, and followers harmonize their bodies to ward off disease. Patients with degenerative diseases such as cancer are promised that the macrobiotic harmony can rid the body of the undesirable illness, even if it has been declared "medically incurable." Certainly, many people with various cancers would be quite willing to eat fermented soybeans and chickpea soup in the hope of complete recovery. How many patients with cancer have we all seen who have adopted the "what-have-I-got-to-lose" attitude?

The book's main danger lies in the prescriptive style enmeshed in a web of misinformation about diet and cancer. The cancer-prevention (i.e., macrobiotic) diet is described, and its adoption is "supported" by "prophetic voices" — historical accounts of ancient provegetarians — and by deceptive statements of "research" summaries, including the National Academy of Sciences' report on diet and cancer, defined as "similar in direction" to macrobiotics. The statement that "hundreds of thousands of people now following the Cancer-Prevention Diet are free of worry from cancer, heart disease, and other degenerative illnesses" is a ploy, totally without supporting evidence, to attract readers who want to be able to control their own destinies. It is also cruel to inform the patient with cancer that "he or she was directly responsible for the development of the disease."

Kushi appears not to be averse to practicing medicine without a license. He advises patients with cancer not to "combine the Cancer-Prevention Diet with radiation or chemotherapy." And patients with advanced cancer are told that with a macrobiotic diet, they are more likely to recover than if they received conventional treatment. Drugs, including insulin, anticonvulsives, and blood pressure medications, are to be eliminated. Disappearance of pain is promised within 10 days after adoption of the diet. And even those without cancer are led to believe that "as many as 80 to 90 per cent of modern people have some type of precancerous condition," such as freckles, dry skin, or sinus problems.

At one point in the book, Kushi advises readers to "use commonsense." Unfortunately, many patients with cancer and a considerable number of those prone to cancer phobia discard common sense in favor of misleading promises and magical cure-alls. The end results of the dangerous persuasions of books such as this one often prove tragic for the patient.

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